



Theodore Chang

Ph.D. (Earthquake Engineering, 2020)

M.Sc. (Distinction) (Civil and Structural Engineering, 2014)

B.Eng. (Civil Engineering, 2013)

>_ Contact

 +49 15510 720730
 github.com/TLCFEM
 tlcfem@gmail.com
 0000-0002-4911-0230

>_ Professional Summary

Computational Mechanics Researcher and Scientific Software Developer with more than a decade of experience specializing in nonlinear finite element analysis (FEA) and high-performance computing. Expert in developing scalable numerical frameworks (C++, Python) for solid mechanics and material science. Proven track record in translating complex mathematical models into production-ready simulation software.

>_ Technical Expertises

-  **Numerical Analysis & Simulation**
Static and dynamic nonlinear finite element analysis of reinforced concrete, steel structures, and continuum mechanics.
-  **High-Performance Software Engineering**
Parallel computing across different architectures using OpenMP, MPI, and SYCL. Development of heterogeneous FEA frameworks and various applications.
-  **Data Engineering & FAIR Principles**
Developing standardized frameworks for scientific data workflows and distributed web-based platforms.
-  **Applied Mathematics**
Linear algebra, differential equations, numerical methods and machine learning topics.

>_ Technical Skills

Languages	C/C++, Python, Fortran, MATLAB, Go, Java, Rust
HPC/DevOps	OpenMP, MPI, SYCL, CMake, Git, Linux, Docker
Analysis	ANSYS, ABAQUS, OpenSees, SAP2000, AutoDesk Products
Editing	L ^A T _E X, Office Suite, Vim

>_ Work/Lecturing Experience

 2026 – present, wissenschaftliche Mitarbeiter  Max-Planck-Institut für Radioastronomie.

 2022 – 2026, wissenschaftliche Mitarbeiter  Humboldt-Universität zu Berlin.

 2016 – 2021, Lecturer, Teaching Assistant (Coordinator)  University of Canterbury.

>_ Referees

 Dr. Chin-Long Lee
Associate Professor, University of Canterbury. chin-long.lee@canterbury.ac.nz

 Dr. Gregory MacRae
Professor, University of Canterbury. gregory.macrae@canterbury.ac.nz

 Dr. Rajesh Dhakal
Professor, University of Canterbury. rajesh.dhakal@canterbury.ac.nz

>_ Projects

- ✎ T. L. Chang, suanPan – an open source, parallel and heterogeneous finite element analysis framework (2021). doi:10.5281/ZENODO.1285221
- ✎ ezp – lightweight C++ wrapper for linear system solvers on HPC clusters
- ✎ motion-base – a web-based ground motion database
- ✎ msglc – (de)serialize json objects with lazy/partial loading containers using msgpack/cbor
- ✎ Damping Dolphin – a utility to visualise and generate parameter sets for use in the damping model
- ✎ Alcongrator – a utility to process and visualise ground motions
- 📖 suanPan-manual – manual for open source FEM platform suanPan
- 📖 structmech – a TikZ command set for structural mechanics drawings

>_ Research

🔖 Research Interests

constitutive models, couple stress theory, finite element formulation, structural dynamics, isogeometric analysis

🔖 Ph.D. Thesis

L. Zhang, Numerical analysis tools for modelling reinforced concrete shear wall buildings subjected to earthquake loading, Ph.D. thesis, University of Canterbury (2020). doi:10.26021/1833

🔖 Books

- 📖 G. MacRae, T. L. Chang, Introduction to structural steel (2021). doi:10.5281/ZENODO.5513880
A textbook on structural steel in accordance with the AU/NZ standards.
- 📖 (WIP) Constitutive Modelling Cookbook
A reference book on constitutive modelling covering formulation and implementation.

🔖 Journal Papers

- 📄 T. Chang, C.-L. Lee, Some alternative forms of nonviscous damping (Sep. 2026). doi:10.31224/8137
- 📄 T. Chang, The balloon-v1 model for metals with improved cyclic response (Apr. 2026). doi:10.31224/6937
- 📄 S. Shabih, L. Pielsticker, F. Dobener, A. Albino, T. Chang, C. Emminger, L. Ginzburg, R. Hildebrandt, M. Kühbach, R. Mozumder, T. Pincelli, M. Aeschlimann, M. Grundman, W. Hetaba, C.-A. Palma, L. Rettig, M. Scheidgen, J. A. M. Prieto, C. Draxl, S. Brockhauser, C. Koch, H. B. Weber, pynxtools: A python framework for generating and validating nexus files in experimental data workflows (Nov. 2025). arXiv:2511.10105, doi:10.48550/ARXIV.2511.10105

-  T. L. Chang, On the subloading surface model for metals: some insights and an efficient numerical implementation, *Acta Mechanica* (May 2025). doi:10.1007/s00707-025-04339-0
-  T. L. Chang, C.-L. Lee, Spurious response due to linear interpolation of input load and some remedies, *Journal of Earthquake Engineering* (2024) 1–22doi:10.1080/13632469.2024.2372814
-  T. L. Chang, C.-L. Lee, A strategy for fast evaluation of nonviscously damped systems with arbitrary kernels, *Mechanical Systems and Signal Processing* 210 (2024) 111156. doi:10.1016/j.ymssp.2024.111156
-  T. L. Chang, C.-L. Lee, Reformulation of concentrated plasticity frame element with N - M interaction and generalized plasticity, *Journal of Structural Engineering* 150 (1) (2024) 04023210. doi:10.1061/JSENDH.STENG-12176
-  M. Scheidgen, L. Himanen, A. N. Ladines, D. Sikter, M. Nakhaee, Á. Fekete, T. Chang, A. Golparvar, J. A. Márquez, S. Brockhauser, S. Brückner, L. M. Ghiringhelli, F. Dietrich, D. Lehmborg, T. Denell, A. Albino, H. Näsström, S. Shabih, F. Dobener, M. Kühbach, R. Mozumder, J. F. Rudzinski, N. Daelman, J. M. Pizarro, M. Kuban, C. Salazar, P. Ondračka, H.-J. Bungartz, C. Draxl, NOMAD: A distributed web-based platform for managing materials science research data, *Journal of Open Source Software* 8 (90) (2023) 5388. doi:10.21105/joss.05388
-  C.-L. Lee, T. L. Chang, A. J. Carr, Consistent assembly method for elemental damping, *Computers & Structures* 289 (2023) 107152. doi:10.1016/j.compstruc.2023.107152
-  W. Dong, M. Li, T. Sullivan, G. MacRae, C.-L. Lee, T. Chang, Direct displacement-based seismic design of glulam frames with buckling restrained braces, *Journal of Earthquake Engineering* (2022) 1–32doi:10.1080/13632469.2022.2110999
-  T. L. Chang, C.-L. Lee, New mixed formulation and mesh dependency of finite elements based on the consistent couple stress theory (Jul. 2022). arXiv:2207.02544
-  T. L. Chang, C.-L. Lee, Numerical simulation of generalised Maxwell-type viscous dampers with an efficient iterative algorithm, *Mechanical Systems and Signal Processing* 170 (2022) 108795. doi:10.1016/j.ymssp.2021.108795
-  T. L. Chang, C.-L. Lee, A. J. Carr, R. P. Dhakal, Numerical evaluations of a novel membrane element in response history analysis of reinforced concrete shear walls, *Engineering Structures* 220 (2020) 110760. doi:10.1016/j.engstruct.2020.110760
-  T. L. Chang, C.-L. Lee, A. J. Carr, R. P. Dhakal, Numerical evaluations of a novel membrane element in simulations of reinforced concrete shear walls, *Engineering Structures* 199 (2019) 109592. doi:10.1016/j.engstruct.2019.109592
-  T. L. Chang, C.-L. Lee, A. J. Carr, R. P. Dhakal, S. Pampanin, A new drilling quadrilateral membrane element with high coarse-mesh accuracy using a modified Hu–Washizu principle, *International Journal for Numerical Methods in Engineering* 119 (7) (2019) 639–660. doi:10.1002/nme.6066
-  M. Arnaiz, T. A. Cochrane, N. F. D. Ward, T. L. Chang, Facilitating universal energy access for developing countries with micro-hydropower: Insights from Nepal, Bolivia, Cambodia and the Philippines, *Energy Research & Social Science* 46 (2018) 356–367. doi:10.1016/j.erss.2018.07.016



Conference Papers

-  C.-L. Lee, T. L. Chang, Implementation and performance of bell-shaped damping model, in: F. Di Trapani, C. Demartino, G. C. Marano, G. Monti (Eds.), Proceedings of the 2022 Eurasian OpenSees Days, Springer Nature Switzerland, Cham, 2023, pp. 147–156.
doi:10.1007/978-3-031-30125-4_13
-  C. Lee, T. Chang, Numerical evaluation of bell-shaped proportional damping model for softening structures, in: 15th World Congress on Computational Mechanics (WCCM-XV) and 8th Asian Pacific Congress on Computational Mechanics (APCOM-VIII), CIMNE, 2022.
doi:10.23967/wccm-apcom.2022.083
-  T. Chang, C.-L. Lee, A. Carr, R. Dhakal, Dynamic analysis of a reinforced concrete shear wall building using a novel finite element, in: Proceedings of the 7th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering (COMPDYN 2019), Institute of Structural Analysis and Antiseismic Research School of Civil Engineering National Technical University of Athens (NTUA) Greece, 2019, pp. 4565–4575.
doi:10.7712/120119.7250.18782